Assay: Business Proposal v2

Nov 1, 2026 · Jan-Albert Viljoen



Pick like a Pro. *Old-world value, with a quant in your pocket.*

Executive summary

Assay is a low-cost investing research app that keeps financial knowledge free and charges for the tools that make it usable. It screens US shares with transparent quantitative models, shows every trade and assumption behind a pick, and tests every model against the S&P 500 total return on data it was never tuned on.

- **The problem.** Quantitative research is expensive, so it mostly serves people with capital. Ordinary investors stop at index funds because picking shares feels like guessing.
- **The philosophy.** No one should accept a pick because an algorithm made it. Education and foundational research stay free; members pay for advanced models, customisation, convenience and infrastructure. *When everybody is mining for gold, sell the shovel.*
- **Three layers.**
  - **Standard** (free): education, community content, limited Newtonian models built on Graham's rules, and the Midas chat assistant to explain the picks.
  - **Plus** (about $4 a month): Einsteinian machine-learning models, full variants and customisation, and wallet tracking that remembers each purchase and plans what to sell, when and how much.
  - **Studio** (up to about $15 a month): the Midas platform where creators build, test, accredit and distribute models, earning a commission when members use them; a reduced fee for members who bring their own AI.
- **The difference.** Reports that show the working, including failed picks, and one testing standard for every model, house or creator. Accreditation certifies method, never returns.
- **The hurdle.** About $50,000 a year in operating profit before tax, as a side business beside the founder's day job, not a venture seeking funding. Under illustrative assumptions that takes about 900 to 2,700 paying members, depending mostly on the price-data licence and billing (Appendix E).
- **Where it stands.** The research engine runs on S&P 500 data. In its current report, value picks bought twice a year and held are well ahead of the S&P 500 for every start year: from 2020,

33.9% a year against 15.9%. The rules were chosen while looking at the same years, only today's index members are scored, and one company holds half the value, so this is a promising research result, not yet proof of a lasting edge.

- **The next step.** A free private beta, to test the method on years it was not built on and to measure demand, conversion, churn and willingness to pay. Data-licensing costs and the regulatory position are settled before any paid launch.

A satellite, not a replacement. Midas complements diversified investing: it gives access to researched, higher-risk opportunities for a smaller part of a portfolio, while the core stays in established indices. It is not suitable for every investor, and the investor decides what fits.

The problem and the opportunity

Sophisticated investment research is expensive to produce, so it is mostly sold to people and institutions with substantial capital. Ordinary investors decide with far less: few transparent methods and little objective evidence of what works.

This is commercially understandable. Research needs expertise, data and infrastructure, and hedge funds, managers and advisers protect it behind subscriptions and client relationships. The practical result is that most retail investors stop at index funds, because picking shares feels like guessing: which companies, how much in each, and when to sell.

- **South African retail investors** hold mostly index funds and ETFs. Platforms such as EasyEquities give them cheap access to US shares in fractions, but no guidance on which shares to buy.
- **Part-time traders** write strategies, sometimes with an AI chat assistant, but cannot tell whether a good backtest was skill or luck. They lack the quant background to test it properly, and no affordable tool does it for them.
- **The target member** sits between the index investor and the day trader: willing to take more risk than the S&P 500, but wanting to look at the portfolio about once a year.

The opportunity is the gap between expensive professional research and free tips with no proof. Graham's rules are simple, published and testable, so screening on them costs little to run. What has been missing is honest evidence and a clear report, delivered cheaply.

Philosophy: the knowledge stays free, the tools are paid

Assay does not sell scarce knowledge. It sells the tools that make knowledge usable. The philosophy matters more than the app: the app is how it is put into practice.

A member should never have to accept a pick because an algorithm produced it. For every model, Assay aims to show:

- why the model chose each company, and which rules or inputs drove it;
- the assumptions, data and method behind it, so an interested member can study and reproduce it;
- its full history, including the picks that failed;

- whether its record could reasonably be explained by chance.

What stays free, and what is paid. Financial education and meaningful access to foundational research stay free; they are not withheld to force a subscription. What members pay for is what costs Assay to build and run:

- more sophisticated models (the Einsteinian line), the full range of variants and customisation, and convenience;
- the infrastructure to develop, test, validate, host and distribute models;
- access to models that independent creators contribute, who earn from their work.

When everybody is mining for gold, sell the shovel. The gold is the knowledge, and Assay leaves it in the open; the shovels are the tools, the testing and the distribution.

Investment philosophy: a satellite, not a replacement

Midas is designed to complement a diversified portfolio, not to replace it. It gives investors transparent, quantitative research into higher-risk opportunities that may form part of a smaller satellite allocation within a broader strategy.

Core and satellite. The founder's own portfolio illustrates the idea:

- **Core:** most of the capital sits in index funds that together cover companies across the world: the United States, developed markets, Europe, the United Kingdom, Japan and emerging markets, with some property and bonds. Contributions fill the tax-free savings account first, then regular accounts at the same weightings.
- **Satellite:** about 5 to 10% of new investments goes to higher-risk opportunities researched with Midas, which may offer higher returns but carry materially more uncertainty and downside risk.
- **Risk containment:** the small allocation limits how much unsuccessful bets can hurt the whole.

The 5 to 10% is the founder's personal choice, shown as an illustration. It is not a recommended allocation, and no single allocation suits every investor.

It follows simple, long-standing principles of financial planning: diversify the core across companies worldwide, invest a fixed amount at regular intervals, buy and hold (the founder has never sold a share), keep speculative positions small and sized to the loss one can bear, and move towards safer assets with age. The split is weighted for a long horizon and is more aggressive than Graham's defensive investor, who holds between 25% and 75% in shares; for general questions, Graham is the reference.

The path with age. Shares (higher risk) against bonds and property (lower risk), by years to retirement: 90/10 at 40 years, 75/25 at 30, 50/50 at 20 and 25/75 at 10. The 40-year end matches Warren Buffett's own instruction for his wife's inheritance, 90% in a low-cost S&P 500 index fund and 10% in short-term government bonds (letter to shareholders, 2013); from 30 years onwards the path stays inside Graham's range. The free Midas chat explains this path to any member who asks, with Graham as the reference for general questions.

Full disclosure. The split above is the founder's listed portfolio. Most of his wealth today is property, through his own asset-management business for residential property and short-term rental bookings in Cape Town, because a house cannot be bought in small regular amounts. The business is set up as a company for its tax benefits, and its returns are moved into the listed split over time, until property is a small part of the whole. A portfolio concentrated in one city and one kind of asset carries its own risks, which is why the plan is to dilute it.

The Midas portfolio split. Plus members can enter their core holdings next to their satellite picks. Midas then shows the whole portfolio: the split between core and satellite, and how the satellite shifts its exposure by sector, country and single company. A member who already holds a technology-heavy index fund sees at once when a pick adds still more technology. The founder's own split is the first worked example. The member sets the split; Midas shows its consequences and never prescribes it.

The role of Midas. A cautious investor may put a small part of a portfolio into cryptocurrency or other speculative assets without letting it replace the investments that build their wealth. Midas offers another way to fill that satellite: quantitative models, documented methods and transparent performance records instead of sentiment or informal tips. The aim is not to remove the risk of higher-risk investing but to give investors better tools to understand, evaluate and manage it.

What Midas is not.

- It does not replace diversified index investing, conventional investment products or professional financial advice.
- It does not suggest moving a whole portfolio into model-selected shares.
- It does not assume that active or algorithmic strategies will consistently beat diversified indices.
- The investor remains responsible for deciding whether an investment suits their circumstances and portfolio.

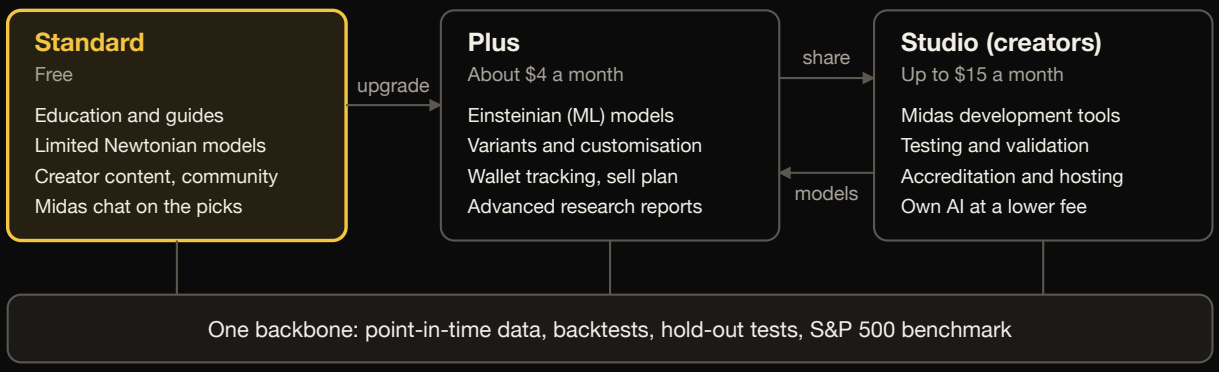
How it connects to transparency. Assay should help members tell apart four different things: a strategy with a credible method; a strategy with attractive past returns; a strategy whose performance has been tested out-of-sample; and a strategy that may simply have benefited from favourable markets or chance. Past performance is never presented as a guarantee of future returns, and members are encouraged to judge a model against their whole portfolio, risk tolerance, horizon and existing exposures.

This is an investment philosophy, not personalised advice or a recommended asset allocation.

Product layers

Assay has three layers with different purposes, not three price points for more of the same: Standard educates and builds the community, Plus sells advanced research, and Studio sells the infrastructure for model creators.

Free knowledge feeds the platform; research and creator tools pay for it



Creators build and accredit models in Studio; Plus members use them; creators earn a share of that revenue.

Assay ecosystem · three product layers on one testing backbone

- **Standard (free).** Part of the philosophy, not a teaser. Members get educational material, limited access to the Newtonian models (Graham's rules), content from creators, the community's knowledge, and the Midas chat assistant to explain the picks. The chat is a cost Assay carries for every free user. Free members manage their own investments: they decide how much to put in each pick and when to sell. The exact limits on Newtonian access are still to be decided.
- **Plus (about $4 a month).** For members who want more sophisticated research and more control. Einsteinian machine-learning models are available only here, with the full range of model variants, input customisation and fuller reports. Plus also tracks the member's wallet: the member enters the exact amount to invest, Assay remembers every pick bought, and it says what to sell, when and how much. Accredited creator models may be included; whether all of them are, or some carry a separate fee, is still to be decided.
- **Studio (up to about $15 a month).** A different product: the Midas development platform for independent researchers. Creators build and configure models, test them on historical data with the same framework the house models use, submit them for accreditation, and have accredited models hosted and offered to Assay's paying members, earning a commission on their use. Assay provides the workspace, testing, hosting and access to members; the creator provides the research and keeps improving the model. Members who connect their own AI subscription, such as Claude, to a restricted Midas server pay a reduced fee, because they carry the AI cost themselves.

Data for Studio members. Studio also opens the cleaned and standardised datasets behind the house models, so creators test on the same data: point-in-time fundamentals, prices and dividends, index membership and the derived metrics. The aim is one of the most thorough market-data platforms for independent researchers, extended over time to developing economies, so strategies can be tested in different market environments, not only the US. Which datasets can be offered depends on their licences (Appendix G).

The creator layer is called Studio, not Premium, because it serves creators rather than investors.

In every layer, Assay never holds money or executes trades: members buy on their own platform and tell Assay what they bought.

What sets Assay apart

The value is not the pick but the evidence behind it: a member can see how a result was reached and judge whether to trust it. The intended advantage is a trusted research environment, not machine learning as such; it is a goal to be earned, not an advantage already held.

- **Transparent reports.** Every trade behind a conclusion, the bad picks beside the good, and the rules or inputs behind each pick (Appendix C). Competitors mostly show a score or a dashboard.
- **One testing standard for every model.** House models and creator models run on the same engine: fundamentals taken as first filed with the SEC, a hold-out period the model never saw, and the S&P 500 total-return index as the benchmark.
- **"Was it luck?" tests.** Planned: comparison with random portfolios of the same kind, and a risk-adjusted return corrected for the number of variants tried.
- **Two model lines, compared like for like.** Newtonian rules any member can read; Einsteinian models for those who want more, judged by the same tests.
- **Free education that is genuinely useful.** The free layer teaches the method rather than teasing it, which gives members a reason to stay and share.
- **An ecosystem that can grow.** Independent researchers can build, accredit and distribute models without building their own platform.
- **Safer in a bubble's correction, by design.** The rules leave out overvalued shares, so a correction of an AI bubble should hurt the Newtonian models less than the index; the cost is trailing while a bubble inflates, and the reports show both phases.
- **Speed and cost.** Large research firms still run on spreadsheets and slow approval processes; an automated report pipeline produces a full report in minutes.
- **A brand unlike the rest.** Competitors look like Bloomberg terminals or crypto apps. Assay looks like a fine mechanical watch: precise, certain, and quietly modern.

Brand and positioning

Assay is old school colliding with fintech: Graham's almost forgotten principles, steered by an AI assistant, offered at small prices with the look of quality.

- **The name.** To assay is to test whether gold is genuine. The word comes from Old French *assai*, a form of *essai* ("trial"), from Late Latin *exagium* ("a weighing"), and shares its root with *essay*: fitting for an app that weighs companies and writes a report on them. Assay offices still stamp the hallmark that certifies a precious metal; Assay's reports are meant to carry the same weight.
- **Midas, the assistant and the platform.** Midas is the chatbot and the development platform: the king with the golden touch who has learned to be careful with it. Midas stands for the infrastructure that lets ordinary people take part in creating wealth, without any promise of returns. A possible line, for approval: *Midas: turning everyday labour into wealth.*
- **The look: fine watchmaking.** In a world of smartwatches, a mechanical watch from a maison such as Patek Philippe still stands for precision and certainty that last generations. Assay is that watch with modern flair, like a fine mechanical watch that can also pay contactlessly. The cues

are the dial, the hand-finished movement behind a glass caseback, and the chronometer certificate that proves its accuracy, as Assay's hold-out tests are meant to prove a strategy. The logo sets Midas's crowned face inside a dial.

- **Long-term value.** Patek Philippe's line, that you never own one but only look after it for the next generation, is the spirit of value investing. Assay borrows the idea, not the words or the marks.
- **A second inspiration.** The hand-built Dutch supercars of Spyker, whose aviation history shows in their turned-aluminium dashboards and exposed gear linkage, add a sense of craft and adventure.
- **A motto.** Seneca's *Ignis aurum probat*: fire tests gold. It names what the platform does, testing every strategy under stress before it is trusted.
- **The counterpoint to crypto.** Everyone chases quick wealth in assets with no measurable value. Assay keeps the spirit of adventure, hunting for underdogs that can become giants, but bases every choice on value that can be measured. It is counting cards, not a lottery.

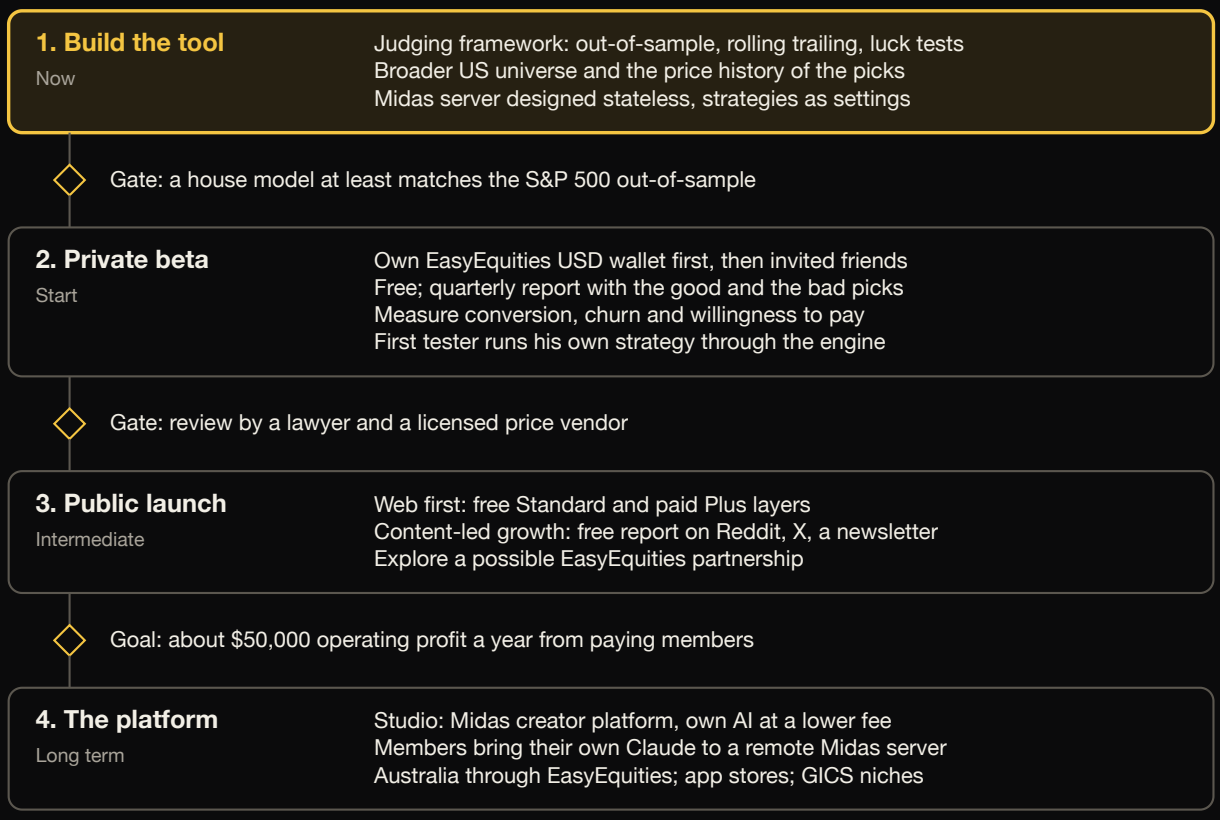
The slogan: *Pick like a Pro.* *Old-world value, with a quant in your pocket.* It joins the promise (choosing shares with a professional's method) to the brand's two halves: Graham's tried and tested rules, and modern quantitative tools. The earlier candidate naming Warren Buffett is dropped.

Midas's character. The same Midas speaks everywhere in Assay: the king who wished everything he touched would turn to gold, starved surrounded by it, and learned that gold is not value. Since washing the touch away in the river Pactolus, he tests before he touches. He is patient, thinks in decades, shows his failed picks first, and admits doubt ("I've been wrong before; I have the ears to prove it," after the donkey's ears Apollo gave him for bad judgement). His teachers are Graham's book and Buffett's method, which he quotes but never imitates. His signature lines: "*Let us assay it first.*" and "*We never go to the moon; we only reach for the stars.*"

Route to market and timeline

Assay grows at the pace of a side business: each phase opens only when the gate before it is passed, not on a fixed date. The first job is to prove demand, retention and willingness to pay before building the creator marketplace.

Assay grows in four phases, each opened by a gate



Roadmap · 4 phases, 3 gates · phases are not to scale

The tool comes first, because the product is the proof. Content comes before the app: the free quarterly report and its public track record build the audience. No paid advertising is planned early, because clicks on finance keywords are expensive.

How free users would find Assay. Each channel is a mechanism to test, not a forecast of growth: the free report shared in value-investing communities; creators bringing their own followers; testers referring friends; and the public track record, which grows more credible every quarter.

EasyEquities and Australia. EasyEquities is where the founder invests, and its USD wallet holds the US shares the screen picks. A partnership is an aim, not an agreement: no partnership exists today. If it happened, Assay would give EasyEquities clients a reason to move beyond index funds, and EasyEquities, as a licensed provider, might help answer the licensing question. EasyEquities also serves clients in Australia, which makes Australia a natural later expansion.

Immediate next steps: the beta

The beta costs the founder almost nothing: invited testers open the research environment in their browser, bring their own Claude subscription, and fetch market data themselves from free public sources. A downloadable app comes only once testers keep coming back.

1. **Finish judging the model.** Test the house model on years it was not built on, with the historical members of the index, so the report's lead is measured fairly.

2. **Make the environment beta-ready.** A one-click browser workspace (GitHub Codespaces), a short quick-start guide, the Midas server registered for Claude, and one command to fetch the data. Check that no keys or vendor data are committed before anyone is invited.
3. **Invite two or three testers** as collaborators: one who wants better picks, one who wants to test his own strategy.
4. **Watch and measure.** Time to a first result, what confused them, which features they used, and what they would pay for.
5. **Decide on the app** from what the testers did, not what they said.

Beta item	Cost to the founder	Note
Private repository with invited collaborators	$0	GitHub's free plan allows private repositories with collaborators
Browser workspace (GitHub Codespaces)	$0	Testers use their own free monthly allowance; if Assay paid instead, roughly $0.18 an hour per 2-core workspace (approximate)
AI assistant	$0	Testers connect their own Claude subscription to the Midas server
Market data	$0	Each tester fetches from free public sources (SEC EDGAR, FRED, Yahoo Finance) for personal use; nothing is redistributed
Domain and landing page	about $15 to $50 a year	Optional during the beta
Downloadable app, later	$99 a year (Apple Developer Program), $25 once (Google Play)	Only when the app is built
Legal review	one-off, to be quoted	Before any paid or public launch, not for the free beta

Codespaces allowances and prices are from general knowledge and should be checked on GitHub's pricing page before the beta starts.

Key risks

The main risks are in Appendix G; in short:

- **Regulation is unresolved.** Not holding money or executing trades does not by itself take Assay outside financial-services rules; a securities lawyer advises before any paid or public launch.
- **The edge is not yet proven.** The report's lead over the S&P 500 is in-sample, with survivorship and concentration caveats.
- **Licensed data is the largest unknown cost** and may not be resold just because it has been reshaped.
- **Conversion and churn** decide whether the hurdle is reachable; the beta measures them.
- **Investment risk.** Higher-risk, model-selected investments can lose a substantial part of their value; a small satellite allocation limits the damage but does not remove it.

Appendix A: About the founder

Jan-Albert Viljoen brings the two worlds Assay joins: the rigour of theoretical physics and the stability that enterprise finance demands.

- **Physics.** A South African theoretical astrophysicist by training, with a PhD in cosmology specialising in general relativity. The Einsteinian model line takes its name from that background.
- **Fintech.** After his doctorate he moved into fintech, training credit-risk and other risk models with machine learning.
- **Big Four.** He is a Financial Risk Manager in the quantitative department of a Big Four firm, where most of his work brings fintech practice to large banks: upgrading their software infrastructure where architecture changes slowly.
- **Now.** He has built an enterprise-grade development environment in which an MCP server helps extend the code base at speed, while the financial logic follows industry standards and the infrastructure stays reliable and auditable.

He also runs an asset-management business for residential property and short-term rental bookings in Cape Town, which holds most of his own wealth. Assay's value screen leaves out the Financials and Real Estate sectors, so the house models do not pick property shares. Assay is built the same way as his work: old-world stability, new-world innovation.

Appendix B: Creators and accreditation

Creators contribute in two different ways with different economics: content, which builds the free community, and models, which earn a commission from paying members. The model marketplace is a longer-term opportunity, not a revenue stream on day one.

Content creators publish commentary, educational material, research, investment ideas, market analysis and explanations of quantitative methods. Publishing stays free of any paywall, so creators help build the audience. How content creators might earn is still to be decided.

Model creators contribute intellectual property. The intended process:

1. Build a model in the Midas development environment (Studio).
2. Test it with the same framework as the house models.
3. Have it evaluated against defined methodological and performance criteria.
4. Submit it for accreditation.
5. Once accredited, it is hosted and offered to paying members.
6. The creator earns an agreed commission from the subscription revenue attributable to the model.

Assay provides	The creator provides
Development, testing and validation tools	The model's intellectual property
Hosting, accreditation and distribution	Research, development and method
Access to paying members; platform upkeep	Ongoing improvement and maintenance

The commission is meant to be generous, because Assay gains from the wider ecosystem, but no rate has been set. Still open: gross or net basis, attribution per model or per member, members who use several models, how long commissions run, which fees come off first, refunds and chargebacks, ownership of the intellectual property, and maintenance duties.

Accreditation is quality control, not a promise of returns. It certifies that a model meets a defined technical and methodological standard. It does not mean the model is expected to beat the market, and it never guarantees positive returns. Criteria under consideration, not all yet built:

- results that can be reproduced;
- point-in-time data with no look-ahead;
- documented assumptions and transaction costs;
- out-of-sample testing against the S&P 500 total return;
- checks for overfitting and selection bias, and a test of whether the record could be chance;
- disclosure of limitations and failure cases.

Appendix C: What an Assay report contains

Every pick comes with the working behind it. The template below shows what a member receives; it holds no figures, because no model has yet earned the right to publish them. The proof-of-concept engine already produces the parts marked *built*; the rest is planned.

Part of the report	What it shows	Status
The pick	Company, date, weight in the portfolio, and the model that chose it	built
Why	Which Graham rules the company passed and failed, or which inputs drove an Einsteinian model	built (Newtonian)
Method and assumptions	Data sources, point-in-time handling, rebalancing, costs	partly built
Testing period	The period the model was built on, and the hold-out period it never saw	built
Benchmark	S&P 500 total return, with Berkshire Hathaway as a second comparison	built
Out-of-sample result	Return against the benchmark in the hold-out period, year by year	built
Hit rate	Share of picks that beat the benchmark, by holding period	built
Every trade	A ledger of each year's picks, their returns, and the bad years beside the good	built
Risk	Drawdown and how often the model trailed the index	planned
Was it luck?	Comparison with random portfolios, and a return corrected for the number of variants tried	planned
Limitations	Where the model has failed and what it cannot see	planned

The engine's current report shows why the report matters. Value picks bought each January and July and held are well ahead of the S&P 500 for every start year; from 2020, 33.9% a year against 15.9%. The same report says what limits that result: the rules were chosen while looking at those years, only today's index members are scored, so companies that failed are missing, and Micron makes up half of the value. A report that shows both the lead and its limits is the product; a pick without them is a tip.

Judging a pick for a satellite allocation. The risk parts of the report (drawdown, how often and how far the model trailed the index, how concentrated its value is, and whether its record could be chance) are there so an investor can judge whether an opportunity suits a limited satellite allocation, and how large a loss on it could be. A concentrated record, such as half the value in one company, is a reason to size the allocation small, not to trust it more.

Appendix D: Competition

Each part of Assay exists somewhere, but no competitor found combines open Graham rules, member-tuned strategies scored out-of-sample, a full written report and a creator platform on one testing standard. Only eToro among the direct competitors is listed; the rest are private.

Competitor	What it offers	Overlap with Assay	Listed
InvestingPro, ProPicks AI	AI strategies updated monthly, the same for every subscriber	Picks and tiers; not personalised, no chat	No
Validea	Model portfolios on gurus' rules, including Graham	Closest on method; no personalisation	No
GuruFocus, Old School Value	Graham Number and guru screens	Graham metrics as a tool for enthusiasts	No
Magnifi	Conversational AI assistant with a brokerage	Chat-steered investing; no stated method	No
Danelfin	AI score from 1 to 10 per stock	Ranked picks; a black box	No
Sterling Stock Picker, StockPick AI	Risk questionnaire, then picks	Close to the Standard layer	No
Simply Wall St, Seeking Alpha	Portfolio tracking, visual reports, quant ratings	Plus-style research and tracking	No
Composer	Build, backtest and automate rule strategies	Testing for technical users	No
eToro, TradingView	Copy trading and paid creators; published scripts and ideas	The creator platform	eToro yes (Nasdaq: ETOR)

Listed neighbours that are not direct competitors: Robinhood (HOOD), Interactive Brokers (IBKR), Charles Schwab (SCHW), Morningstar (MORN), Value Line (VALU), and the data firms S&P Global, MSCI, Moody's and FactSet. The competitor list comes from a web search on 1 November 2026 and is not exhaustive.

Appendix E: Commercial model

Under illustrative assumptions, about 900 to 2,700 paying members reach the hurdle of $50,000 a year in operating profit before tax. The spread comes mostly from two lines: the price-data licence and how members are billed. Keeping members matters as much as finding them: at 4% monthly churn, about 720 payers must be replaced each year. The $50,000 is a test of whether a side business justifies the founder's time, not a fundraising or valuation target; whether it is counted before or after reinvestment and one-off development costs is still to be decided.

Users are not payers. The free layer is deliberate, so most users will never pay. The model separates free users, Plus subscribers and Studio creators, and every figure counts paying customers only.

Contribution per payer. These inputs come from v1:

- **Plus:** $4 a month is $48 a year. What is left depends on billing: $32.80 through an app store (15% fee, $8 of other costs, the v1 assumption), about $41 on monthly card billing, about $45 on annual billing.
- **Studio:** $15 a month leaves $145 to $173 a year on the same costs. The reduced fee for members who bring their own AI is not yet priced.
- **Free users cost a little.** The Midas chat on Claude Haiku 5.5 costs about $0.0005 a question; in the base case the free users behind each payer cost about $0.27 a year.
- **Commissions:** none are included, because no rate has been set and the marketplace comes later. Each 10% of Plus net revenue paid to creators would cost about $4 per Plus subscriber a year.

Scenario (illustrative)	Conservative	Base	Upside
Billing	App store, monthly (15%)	Card, monthly (about 10.4%)	Card, annual (about 3.5%)
Infrastructure a year (Appendix F)	$6,000	$4,000	$1,500
Price-data licence a year	$30,000	$10,000	$3,000
Studio creators	0	25	100
Download-to-paid conversion	1.5%	2.2%	3.0%
Midas chat per free user a year	$0.02	$0.006	$0.001
Monthly churn of payers	6%	4%	3%
Plus subscribers needed	**2,732**	**1,474**	**823**
Total paying customers	2,732	1,499	923
Free users	179,402	66,638	29,844
Annual revenue	$131,136	$75,252	$57,504
Downloads a year to replace churn	131,136	32,706	11,076

Only the prices, the published fee and model rates, and the 2.2% base conversion come from sources; the 2.2% is RevenueCat's median download-to-paid rate for freemium apps (RevenueCat, 2025). Everything else is a placeholder to validate in the beta. The editable model is `docs/assay_unit_economics_v1.xlsx`.

Corrections to v1. At $32.80, 1,500 payers give $49,200 and 2,000 give $65,600, and both are contribution before fixed costs, not profit. v1 also applied a download-to-paid rate to sign-ups and rounded it to 2%; at 2.2% of downloads, 1,500 to 2,000 payers need about 68,182 to 90,909 downloads, not 75,000 to 100,000 sign-ups.

The smallest viable Assay has the Standard and Plus layers, no marketplace, annual billing, a small model for the chat, a modest data licence and low churn. The conservative case shows the risk: an expensive data licence and app-store billing nearly double the members needed. The beta's job is to measure what this table guesses: conversion, churn, willingness to pay $4, and the real price of licensed data.

What listed neighbours earn (latest fiscal year, from results releases):

Company	Revenue	Net income	Net margin
Robinhood (2025)	$4.5 bn	$1.9 bn	about 42%
FactSet (FY Sep 2024 to Aug 2025)	$2.32 bn	$597 m	about 26%
eToro (2025)	$868 m (net contribution)	$216 m	about 25%
Morningstar (2025)	$2.4 bn	$374 m	about 16%
Value Line (FY May 2025 to Apr 2026)	$33.4 m (publishing)	$21.6 m, mostly investment income	operating about 12%

Robinhood Gold earned $50 million in the fourth quarter of 2025 from 4.2 million subscribers, about $4 a subscriber a month, so the price point is proven at scale. Value Line, the closest old-school value publisher, is shrinking: publishing revenue fell 4.7% in its last year. Old-world value sold the old way does not grow; the modern tools have to carry it.

Appendix F: Running costs and infrastructure

At about 2,000 paying and 80,000 free members, infrastructure costs a few thousand dollars a year; the real costs are the licensed price data, payment fees and the founder's time. Figures are ballpark estimates from general knowledge, not quotes, except the Claude prices.

Item	Free data (phase 1)	Paid data (phase 2)	Note
App hosting and API	$300 to $1,200	same	One managed container service or server
Database (members, wallets, holdings)	$200 to $600	same	Managed Postgres
Data warehouse	under $100	$100 to $500	A few GB of files on cloud storage, queried with DuckDB; no warehouse product needed
Scheduled updates and members' backtests	$300 to $1,800	more with emerging markets	Backtests capped per member
Sign-in, email, monitoring, backups	$200 to $1,500	same	Free tiers at first
Midas chat (Claude Haiku 5.5)	about $300 to $600	same	See below
Licensed price data	$0 (beta only)	to be quoted, roughly $3,000 to $30,000 or more	Unavoidable before a paid launch
Infrastructure total	**about $1,500 to $6,000**	**about $5,000 to $40,000**	
Payment fees (not infrastructure)	$3,000 to $9,000	same	See below

- **The chat is cheap on a small model.** Claude Haiku 5.5 costs $0.10 per million input tokens and $0.50 per million output, about $0.0005 a question. Every member sees the same picks, so each pick's explanation can be written once a quarter and shared. A larger model costs about 20 times as much.
- **Annual billing saves more than hosting costs.** A card fee of roughly 2.9% plus $0.30 a charge takes about a tenth of a $4 monthly payment, but only about 3.5% of a $48 annual one.
- **At 2,000 paying members, the base case of Appendix E earns about $70,000 a year in operating profit** after fees, chat, $4,000 of infrastructure and a $10,000 data licence.

Keeping the scope manageable. Specialised competitors do one thing; Assay must not become five companies at once.

- **One engine, many windows.** Data, backtests and reports are one system; free picks, Plus, Studio and the datasets are views of it.

- **Buy, do not build:** sign-in, payments and hosting are bought services. Creators link video from YouTube rather than Assay hosting it.
- **No code sandboxes at first.** Creator strategies are settings the engine runs, not uploaded code.
- **Phase it:** Standard and Plus first; then public and derived datasets; then Studio and the marketplace when creators ask for it; then emerging markets one at a time. Each phase starts when the one before pays for itself.

Appendix G: Risks, regulation and data licensing

Regulation is the largest open risk and is not resolved. Not holding money or executing trades is central to the design, but it does not by itself take Assay outside financial-services regulation, and no disclaimer alone settles it. A securities lawyer's advice comes before any paid or public launch.

Investment risk. Higher-risk, model-selected investments can lose a substantial part of their value, and a few concentrated holdings can fall far further than an index. Keeping them to a small satellite allocation limits the damage but does not remove it, and diversification does not eliminate investment risk. No model, accredited or not, guarantees a return.

Regulatory questions by jurisdiction. In each jurisdiction, the same questions need answers: does publishing research create obligations; are personalised picks treated differently from general research; do the marketplace, accreditation or creator commissions change the position; could any activity count as financial advice; and do the rules depend on where the member lives?

Jurisdiction	Regulator	Why it is on the list
South Africa	FSCA (FAIS Act)	The founder's home market and the first members
Mauritius	Financial Services Commission	A possible place to register; may not cover South African members
Abu Dhabi	ADGM Financial Services Regulatory Authority	A possible place to register
Australia	ASIC (Australian Financial Services Licence)	A later market through EasyEquities; general advice is regulated more lightly than personal advice

Data licensing is a separate cost and dependency that could change the unit economics. Data held for internal research cannot be assumed redistributable. Each use needs its own check: historical prices, real-time or delayed prices, showing data to members, use in backtests and machine-learning models, showing company information and derived metrics, and hosting creator models built on that data. Yahoo price data may not be redistributed in a product, free or paid; SEC EDGAR fundamentals may be used; using the S&P 500 name commercially may need a licence.

Reselling licensed data after reshaping it. Cleaning and reformatting data bought from a vendor does not by itself give a right to resell it. The vendor's contract decides, and most retail and commercial plans forbid redistribution in any form (Tiingo's terms, for example, require a separate agreement to show data in an app). Copyright is a weaker limit than the contract: in the US, facts are not protected and a compilation only for its original selection or arrangement (Feist v. Rural, 1991), but the EU adds a database right against extracting a substantial part of a database, and exchanges license their data under their own redistribution and derived-data policies. The realistic routes are:

- **Public sources** that may be redistributed, such as SEC EDGAR filings, offered freely and in full.
- **A redistribution licence** bought from a vendor or exchange for the datasets Studio members download.
- **Derived data**: scores, ranks and signals from which the original prices cannot be rebuilt. Many licences allow these or license them separately; this is the safest form for data Assay builds on.

- **Bring your own data key**: members connect their own vendor subscription, so Assay processes data it never redistributes; Tiingo's terms name this case explicitly.

Developing-economy data will usually come from the local exchange or a regional vendor, each with its own terms. Which route each dataset takes is a question for a lawyer reading the actual contracts.

Risk	Why it matters	Response
Unproven edge	The report's lead over the S&P 500 is in-sample: rules chosen on the same years, only today's index members, half the value in one company.	Nothing is marketed as beating the market; the public track record starts with the beta.
Overfitting	Members tuning settings, and thousands of creator models, produce winners by luck.	Hold-out scoring for every setting; rankings and creator pay based on results after publication.
Einsteinian models overfit more easily	Machine-learning models have many more settings than Graham's rules.	The same hold-out gate; not published until they pass it.
Accreditation misread as a promise	Members may read "accredited" as "will make money".	Accreditation certifies method, not returns, and the reports say so.
Marketplace needs two sides	It needs credible creators and paying demand at once.	Treated as a later phase; the business must work without it.
Platform liability	Creators paid for models others follow may be giving advice themselves.	Terms and licence cover creators; part of the legal review.
Value lags in bubbles	Value strategies trailed badly in 1998 to 2000 and 2019 to 2021.	Reports show trailing periods openly; the promise is "not worse" over time, not every year.
Churn	At 4% a month, about half the paying base must be replaced each year.	Measured in the beta; the free layer and creators are the retention levers to test.

The private beta is free and limited to invited testers, which keeps it informal; the questions above apply once anyone pays or the app opens to the public.